BUSINESS INVESTIGATION REPORT · NO. 005

UNDERSTANDING ECOSYSTEM STRATEGY

Synopsys and the Architecture of Mutual Dependence

Why can't great companies succeed alone? An investigation into how Synopsys became indispensable to NVIDIA, Apple, TSMC, Intel, and nearly every advanced chipmaker on earth — and what it teaches every business about winning through ecosystems instead of independence.

SUBJECT COMPANY
Synopsys, Inc.

FRAMEWORK
Ecosystem Strategy

SECTOR
Electronic Design Automation (EDA)

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READING GUIDE

Statements are labelled FACT where they describe verifiable, publicly known information about Synopsys, and ANALYSIS where they represent an interpretation applying ecosystem-strategy thinking to that information. See the methodology note on the closing page for sourcing guidance.

01 What Is an Ecosystem Strategy?

FOUNDATIONAL CONCEPT

An Ecosystem Strategy is a business strategy in which multiple independent organizations create more value together than any one company could create alone. Each participant specializes in a different capability; instead of competing everywhere, they cooperate where cooperation increases the value of the whole system.

TRADITIONAL VALUE CHAIN

One company designs, manufactures, sells, and supports its product nearly end-to-end.

ECOSYSTEM MODEL

Arm → Synopsys → Cadence → ASML → Carl Zeiss SMT → KLA → Applied Materials → Lam Research → TSMC → Apple / NVIDIA / AMD — each company performs one specialized role.

Why Synopsys Matters

Modern chips contain billions of transistors, thousands of design rules, and millions of logic gates — far beyond what any engineering team could design by hand. Electronic Design Automation (EDA) software has become essential; without it, modern semiconductor design would be practically impossible.

FACT Synopsys was founded in 1986 and is headquartered in Sunnyvale, California. It does not manufacture chips — it sells the EDA software, semiconductor IP, and verification tools that let engineers turn ideas into manufacturable silicon.

02 The Semiconductor Ecosystem Map

EVERY STAGE, A DIFFERENT SPECIALIST

No single company performs every stage of building an advanced chip. Instead, each stage is owned by whichever company has spent decades becoming world-class at it.

Stage	Company
CPU Architecture	Arm
Chip Design	NVIDIA, Apple, AMD
Design Software & Verification	Synopsys
Manufacturing	TSMC
Lithography	ASML
Optics	Carl Zeiss SMT
Metrology	KLA
Manufacturing Equipment	Applied Materials
Etching	Lam Research
Packaging	ASE

ANALYSIS Every participant in this map depends on the others. Remove Synopsys and chip designers lose the tools to manage billions of transistors; remove TSMC and none of those designs become silicon. The ecosystem functions only because no single company tries to own every row.

03 Synopsys' Core Capabilities

FOUR PILLARS OF THE BUSINESS

1 · Electronic Design Automation (EDA)

EDA software lets engineers design chips, simulate circuits, verify functionality, optimize performance, and prepare designs for manufacturing — the core toolkit that makes managing modern chip complexity possible at all.

2 · Semiconductor IP

Synopsys licenses reusable IP blocks — USB, PCIe, DDR memory interfaces, Ethernet, security modules — so customers integrate proven building blocks instead of redesigning them from scratch, cutting development time, engineering cost, and design risk.

3 · Verification

Verification determines whether a chip will function correctly before it's manufactured. Since a manufacturing error can cost millions of dollars once it reaches silicon, verification software is one of the highest-leverage tools in the entire design flow.

4 · AI-Assisted Chip Design

Newer Synopsys tools use AI to optimize chip layouts, reduce power consumption, improve routing, and accelerate engineering productivity — an increasingly important capability as design complexity keeps outpacing what human engineers can optimize manually.

04 Why Great Companies Depend on Synopsys

EVEN INDUSTRY LEADERS RELY ON IT

— Apple — needs EDA tools to design Apple Silicon processors.

— NVIDIA — uses EDA software to design increasingly complex AI GPUs.

— AMD — uses EDA software for CPU and GPU development.

— Qualcomm — uses EDA software for mobile SoCs.

— Broadcom — uses Synopsys tools across its networking products.

— Intel — uses advanced design automation despite maintaining deep internal engineering expertise.

— TSMC — collaborates with Synopsys to ensure design tools support each new manufacturing node.

Strategic Partnerships

Partner	Purpose
TSMC	Process Design Kits (PDKs) and design enablement
Samsung / Intel Foundry	Foundry-ready design flows and advanced-node support
Arm	IP compatibility
NVIDIA	AI infrastructure and design optimization
Microsoft Azure / AWS / Google Cloud	Scalable, cloud-based EDA workloads

05 Why Ecosystems Win

THE CASE AGAINST DOING IT ALL YOURSELF

Imagine every semiconductor company built its own EDA software, its own lithography machines, its own optics, and its own inspection equipment. Innovation would slow dramatically — each company would be re-solving problems its neighbors already solved better.

Instead, specialization lets every participant become world-class in one area, and the ecosystem advances faster as a whole than any single vertically integrated company ever could.

> **THE KNOWLEDGE NETWORK**
> Universities → Research Labs → Chip Designers → Synopsys → TSMC → ASML → Carl Zeiss SMT → KLA → Customers — knowledge flows in every direction, and innovation becomes collective rather than proprietary to any one firm.

Analysis

ANALYSIS Competitive advantage increasingly belongs to ecosystems rather than isolated firms. Synopsys does not create value by replacing its customers — it increases the productivity of the entire ecosystem. Every improvement in its software enables faster chip design, lower engineering costs, fewer manufacturing errors, and quicker adoption of new process nodes. As semiconductor complexity grows, dependence on specialized ecosystem partners grows with it — the result is mutual dependence, not independence.

06 Business Principles & Comparison

LESSONS BEYOND SEMICONDUCTORS

— Great companies specialize — they do not attempt to master every capability.

— Specialization increases ecosystem productivity for every participant, not just the specialist.

— The value of an ecosystem exceeds the sum of its individual participants.

— Knowledge shared across trusted partners accelerates innovation for the whole system.

— The strongest competitive advantages often belong to companies that enable others to succeed.

Lessons for Entrepreneurs

Most founders ask, "How can I beat everyone?" The better question is: "How can I become indispensable to everyone?" Businesses that provide essential tools, infrastructure, platforms, or services often prove more durable than businesses selling end products alone.

Comparison with Previous Investigations

BI	Concept	Core Insight
001	Value Chain	Value is created through coordinated activities
002	Capability Building	Capabilities compound through long-term investment
003	Economic Moat	Durable advantages arise from hard-to-replicate assets
004	Bottlenecks	Controlling the critical constraint gives strategic influence
005	Ecosystem Strategy	Innovation increasingly occurs through networks, not isolated firms

07 Conclusion & Methodology

WHY CAN'T GREAT COMPANIES SUCCEED ALONE?

Synopsys illustrates that the future of competition is increasingly ecosystem competition rather than company competition. As semiconductor technology grows more complex, no organization — not even NVIDIA, Apple, TSMC, or Intel — can innovate independently. Each relies on a network of specialized partners, every one contributing a capability that strengthens the whole system.

KEY TAKEAWAY

Sustainable competitive advantage is no longer achieved by owning every activity in the value chain, but by becoming indispensable within an ecosystem. Companies that provide essential capabilities, build trusted partnerships, and enable others to innovate are often the ones that shape an industry's future.

BUSINESS PRINCIPLE

The strongest competitive advantages often belong to companies that make everyone else more productive, rather than companies that try to do everything themselves.

Methodology & Evidence Standards

Statements are labeled FACT where they describe verifiable, publicly discussed information about Synopsys (e.g., founding date, business model, named partnerships), and ANALYSIS where they apply ecosystem-strategy thinking to interpret why that fact matters strategically.

Recommended verification sources: Synopsys Annual Reports and investor presentations; Synopsys Design Platform documentation; TSMC's Open Innovation Platform (OIP) and Technology Symposium materials; Arm ecosystem documentation; IEEE publications on Electronic Design Automation; and Semiconductor Industry Association (SIA) reports.

Note: Named partnerships and product details should be re-verified against Synopsys' latest public materials at the time of use, as products, partners, and offerings evolve.